

Personal Learning Journey

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IT3091 Machine Learning | Group 2026 DS 10
Customer Feature Engineering and Transformation

Project context. Our group analysed the UCI Online Retail dataset to identify customer segments and segment-specific product associations for retention, reactivation and cross-selling decisions. This reflection explains my individual contribution, learning and development during the project.

My contribution

My responsibility was to convert cleaned transaction lines into meaningful customer-level features for clustering. I used a fixed analysis date of 10 December 2011 and created Recency, Frequency and Monetary measures for all 4,334 identified customers. Frequency was based on distinct purchase invoices rather than transaction rows. I also engineered TotalQuantity, ProductDiversity, AverageOrderValue, AverageBasketSize, TenureDays and AverageDaysBetweenPurchases. I prepared both a core RFM feature set and an enhanced candidate containing ProductDiversity and AverageBasketSize, then produced feature dictionaries, quality summaries, transformation evidence, manifests and validated handover datasets.

What I learned

This task showed me that feature engineering is a modelling decision, not only a coding step. Two variables may be individually useful but harmful when they give the same behaviour excessive weight. Monetary and TotalQuantity had a Pearson correlation of approximately 0.92, while Frequency and TenureDays had a Spearman correlation near 0.90. I therefore retained several measures for interpretation but did not place all of them directly into clustering. I also learned that missing purchase intervals for 1,505 one-purchase customers were structurally meaningful rather than random errors that should automatically be imputed.

Challenge and judgement

Strong right skew was the main technical challenge. Monetary skewness was about 19.58 and ProductDiversity skewness was about 6.92 because a small number of genuine high-value customers purchased at wholesale scale. Removing these customers would have changed the business population, while raw values could dominate distance calculations. I created log1p versions and preserved raw versions so the clustering member could compare both representations with StandardScaler and RobustScaler. The transformation reduced Monetary skewness to about 0.40 and ProductDiversity to about -0.24. This evidence helped me understand why transformations must be evaluated, documented and linked to model behaviour.

Teamwork and future development

I worked between preprocessing and modelling, so I had to understand both stages. I checked that every customer had complete RFM values, explained feature definitions and transferred unscaled raw and log datasets so scaling could occur inside each model experiment. Peer review helped us avoid premature commitment to one feature set. AI assistance was used for debugging and wording, but decisions were checked through correlations, distributions, assertions and reload tests. In future, I would explore cohort-normalised features, temporal purchase trends and out-of-time feature validation. I learned that strong features make complex models easier to interpret and that a careful handover is part of professional data science.